

7. (a) Crude oil can be separated into simpler mixtures, called fractions. Fractions contain hydrocarbon compounds called alkanes.

The table shows information about some of the fractions obtained from crude oil.

| Fraction       | Boiling point range (°C) | Number of carbon atoms in the alkanes |
|----------------|--------------------------|---------------------------------------|
| refinery gases | –160 to 20               | 1–4                                   |
| petrol         | 20 to 240                | 4–12                                  |
| naphtha        | 100 to 250               | 7–14                                  |
| kerosene       | 200 to 280               | 11–15                                 |
| diesel oil     | 280 to 350               | 15–19                                 |

Use only the information in the table to answer parts (i)–(iii).

- (i) Complete the sentence below by underlining the correct word(s). [1]

As the number of carbon atoms in an alkane increases,  
the boiling point ( **increases** / **decreases** / **stays the same** ).

- (ii) Hexane has a boiling point of 69 °C. [1]  
Give the name of the fraction that contains hexane.

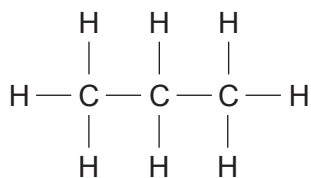
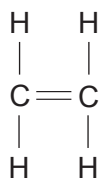
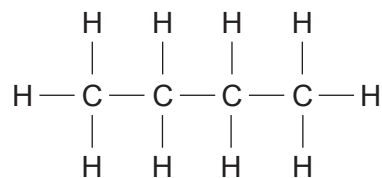
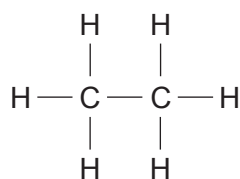
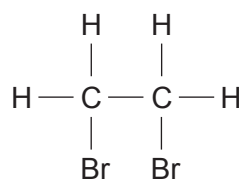
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- (iii) One alkane is found in the kerosene and diesel oil fractions. [1]  
Give the number of carbon atoms in this alkane.

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(b) Diagrams **A–E** show the structural formulae of some carbon compounds.

**A****B****C****D****E**

(i) Circle the molecular formula for compound **A**. [1]



(ii) Give the **letter** of the compound that is unsaturated. [1]

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(iii) Explain why compound **E** is **not** a hydrocarbon. [1]

.....  
 .....

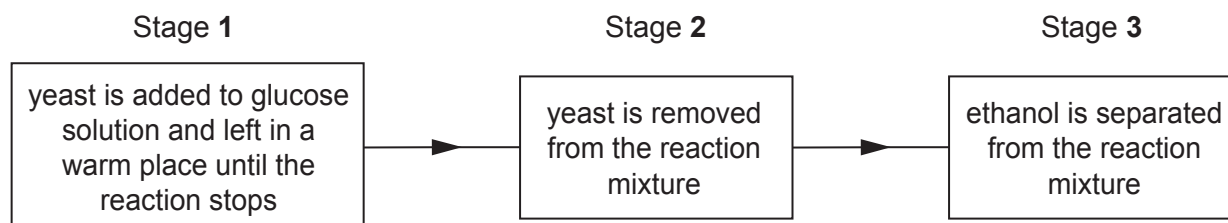
(iv) Give the **letter** of the compound that belongs to the family of carbon compounds with the general formula C<sub>n</sub>H<sub>2n</sub>. [1]

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- (c) The chart shows the stages used in the laboratory to make ethanol from glucose solution.

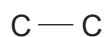


- (i) Give the number of the stage in which distillation is used. [1]

Stage .....

- (ii) The formula of ethanol is  $\text{C}_2\text{H}_5\text{OH}$ .

Complete the structure of ethanol by adding all the atoms and bonds present. [1]



- (iii) Ethanol is the alcohol found in alcoholic drinks.

The following information appears on a bottle of wine.

|                   |                 |
|-------------------|-----------------|
| Volume            | 750 millilitres |
| Alcohol by volume | 13.5%           |

The formula below can be used to calculate the mass of ethanol in an alcoholic drink.

$$\text{mass of ethanol (g)} = 8 \times \text{volume (litres)} \times \text{alcohol by volume (\%)}$$

Use the formula to find the mass of ethanol in a bottle of wine.

[2]

$$1 \text{ litre} = 1000 \text{ millilitres}$$

Mass = ..... g

11

